

Farzad Avishan

Operations Research Scientist | Data Scientist | Optimization & Predictive Modeling
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PROFESSIONAL SUMMARY

Operations research and data scientist with 3+ years of experience building optimization, machine learning, and predictive modeling solutions for complex decision-making problems. Combines a Ph.D. foundation in robust optimization with applied experience in airline, logistics, supply chain, production planning, and quality-control environments. Holds an open Canadian work permit valid until February 2028.

CORE STRENGTHS & TECHNICAL SKILLS

Methods: optimization modeling, decision-support algorithms, predictive modeling, forecasting, exploratory data analysis, model validation, robust optimization, heuristic algorithms, combinatorial optimization, large-scale networks.

Domains and tools: airline operations, logistics, supply chain, production planning, quality analytics; Python, C++, Gurobi, FICO Xpress, CPLEX, GAMS, MATLAB, GitHub/Git; AI-assisted development with Cursor, GitHub Copilot, and agent-based coding workflows.

PROFESSIONAL EXPERIENCE

Research Scientist, Data & AI | Air Canada, Canada | Oct 2024 - Present

- Apply optimization techniques to support business decisions and improve operational efficiency.
- Contributed to a network-planning project that significantly improved operational KPIs and identified substantial cost-saving opportunities.
- Establish repeatable processes for large-scale data analysis, model development, validation, and implementation.
- Conduct exploratory data analysis to identify trends, patterns, relationships, and opportunities in operational data.
- Collaborate with data engineers, machine learning engineers, and other data professionals to support pipeline and model integration.
- Use AI-assisted development tools, including Cursor and GitHub Copilot, to create coding agents, reusable skills, and productivity workflows.

Postdoctoral Research Fellow | HEC Montreal, Canada | 2023 - Oct 2024

- Developed optimization models and solution approaches for applied research and industry-facing problems.
- Conducted research in operations research and optimization, contributing findings to scholarly publications.
- Provided consultative analytical support to industry and private-sector stakeholders.

Research Assistant | Ozyegin University, Istanbul, Turkey | 2021 - 2023

- Awarded Best Research Scientist of the Year for 2022-2023.
- Designed optimization models and solution approaches for production planning, logistics, and supply chain applications.
- Performed literature reviews, computational research, and analytical experiments to support academic studies.
- Compiled and analyzed research results and contributed to scholarly publications and research documentation.

System Analysis Expert | Dadash Baradar Industrial Co., Tabriz, Iran | 2016 - 2017

- Developed predictive models to forecast demand at annual, monthly, and weekly planning levels.
- Managed a production-line setup project, coordinating analytical and operational requirements.

Quality Control Expert | Hadid Parlakh Co., Tabriz, Iran | 2011 - 2015

- Applied statistical process control techniques to monitor production quality and identify process variation.
- Built dashboards to track quality metrics, surface trends, and highlight improvement opportunities.
- Identified out-of-range processes and equipment requiring calibration.

CONSULTING PROJECT

Transportation Cost Forecasting Project | Clear D3, Canada | 2023

- Used historical data to develop predictive models for forecasting transportation costs in future periods.
- Produced model outputs that supported more informed customer price negotiations and pricing strategy decisions.
- Helped improve the company's ability to anticipate cost changes and maintain competitive pricing.

EDUCATION

Ph.D., Industrial Engineering | Ozyegin University, Istanbul, Turkey | Received Mar 2023

- Research field: optimization under uncertainty for production planning, supply chain, and logistics problems.
- Thesis: Applications of Robust Optimization in Logistics and Production Planning. CGPA: 3.87/4.00.

M.Sc., Industrial Engineering | Mazandaran University of Science and Technology, Babol, Iran | Received Sep 2014

- Thesis: A New Optimization Model for Production and Distribution of Supply Chain with Green Logistic Approach.

PUBLICATIONS

Selected papers are available on Google Scholar: <https://scholar.google.ca/citations?user=WfsfoxwAAAAJ&hl=en>